TABLE S11

B004 PARTITION GRANULARITY SENSITIVITY

Clustering	Occupied clusters	Purity	ARI	NMI
Leiden	100	0.6792	0.3168	0.5485
Leiden	200	0.7036	0.2760	0.5363
Leiden	300	0.7149	0.1765	0.5224
FlowSOM	100	0.4958	0.1837	0.4040
FlowSOM	200	0.5035	0.1806	0.3999
FlowSOM	299	0.5071	0.1807	0.3997
SpatialSort	100	0.4758	0.2466	0.3964
SpatialSort	200	0.4910	0.1750	0.3953
SpatialSort	229	0.4955	0.1105	0.3841
PIXIE	100	0.5554	0.2200	0.4214
PIXIE	200	0.5609	0.2122	0.4115
PIXIE	300	0.5641	0.2093	0.4085

K100 and K200 are cell-count-weighted Ward coarsenings of frozen full-precision cluster profiles, sealed before reference scoring. They are not native-K refits or LLM ranking-stability experiments. Metrics average the two test regions equally.